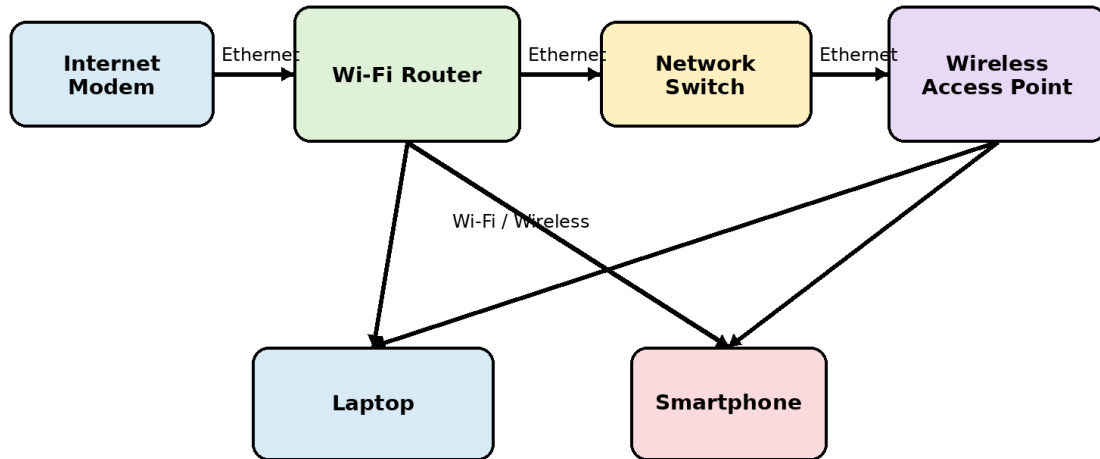


Assignment:-1

Task 1:- Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.

Typical Home Wi-Fi Network Setup



The router manages the home network; the switch provides wired expansion; the access point extends wireless coverage.

Task 2:- List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

Networking Device	Main Function	Real-Life Example
Modem	Connects your local network to the Internet Service Provider (ISP) and converts signals for internet communication.	Used at home to receive an internet connection from a fiber/cable provider.
Router	Directs network traffic between devices and networks; shares one internet connection among multiple devices.	At home to share internet with laptops, smartphones, and smart TVs.
Switch	Connects multiple wired devices within the same local area network (LAN).	Used in an office to connect computers, printers, and servers using Ethernet cables.
Access Point (AP)	Provides wireless Wi-Fi connectivity to devices and extends wireless network coverage.	Installed in a school or office to provide Wi-Fi in different rooms.
Network Cable	Carries data between networking devices using a wired connection.	Ethernet cable connecting a desktop PC to a router or switch.

Task 3:- Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly use

1. Ethernet Cable (Twisted Pair / Cat5e / Cat6)

Common use: Used to connect computers, routers, switches, printers, and other devices in home and office LANs.

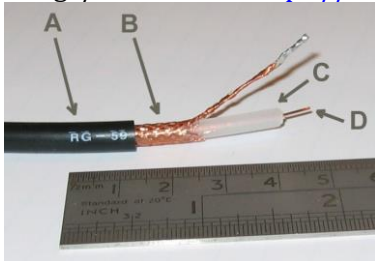
Image/source link: https://en.wikipedia.org/wiki/Category_5_cable



2. Coaxial Cable

Common use: Commonly used for cable television, cable-modem internet connections, CCTV, and radio-frequency signals.

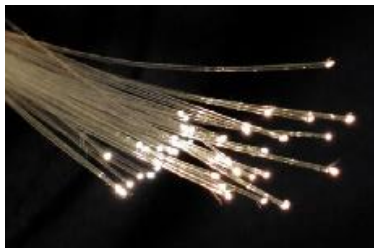
Image/source link: https://en.wikipedia.org/wiki/Coaxial_cable



3. Fiber Optic Cable

Common use: Used for high-speed internet, long-distance telecommunications, data centers, and network backbones.

Image/source link: https://en.wikipedia.org/wiki/Optical_fiber



Task 4:- Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Basis	Router	Switch
Main Use	Connects different networks.	Connects devices in one network.
Typical Location	Used at home or office for internet sharing.	Used in offices, labs, and server rooms.
Data Handling	Sends data between networks using IP addresses.	Sends data to the correct device using MAC addresses.
Internet Connection	Connects the local network to the internet.	Does not directly connect to the internet.
Example	Home Wi-Fi router.	Office network switch.